

JAVA FULL STACK DEVELOPMENT – 3 MONTHS COURSE SYLLABUS

WEEK 1 – CORE JAVA FUNDAMENTALS

- Introduction to Java
- Features of Java
- JDK, JRE, JVM
- Java Program Execution Flow
- Installing JDK and Environment Setup
- Classpath and Extensions Visible Setting
- Writing, Compiling, and Executing Java Programs
- Comments in Java
- Data Types
- Variables
- Operators
- Input and Output using Scanner

WEEK 2 – CONTROL FLOW AND ARRAYS

- Conditional Statements (if, switch)
- Looping Statements (for, while, do-while)
- One-Dimensional Arrays
- Two-Dimensional Arrays
- Command Line Arguments

WEEK 3 – STRINGS AND OOP

- String
- StringBuffer
- StringBuilder
- Class and Object
- Methods
- Constructors
- Inheritance
- Types of Inheritance
- super Keyword

WEEK 4 – ADVANCED CORE JAVA

- Polymorphism
- Abstraction
- Abstract Classes
- Interfaces
- Packages
- Access Modifiers
- final Keyword
- Exception Handling
- Multithreading
- File Handling
- Wrapper Classes
- Collections Framework (List, Set, Queue, Map)
- Java 8 Lambda Expressions
- Regular Expressions

WEEK 5 – MYSQL DATABASE

- DBMS Concepts
- Database and Table Creation
- Data Types
- Constraints
- DDL Commands
- DML Commands
- DQL
- Joins
- Transactions

WEEK 6 – JDBC WITH MYSQL

- JDBC Architecture
- JDBC Drivers
- MySQL Connector
- CRUD Operations using JDBC
- PreparedStatement
- ResultSet
- SQL Injection

WEEK 7 – SERVLETS AND JSP

- Servlet Lifecycle
- Request and Response
- Form Handling
- JDBC Integration in Servlets
- JSP Elements
- JSP Directives
- Action Tags
- Implicit Objects
- JavaBeans
- MVC Pattern

WEEK 8 – FRONTEND TECHNOLOGIES

- HTML Forms and Tables
- CSS Styling and Box Model
- JavaScript Fundamentals
- DOM Manipulation
- Form Validation

WEEK 9 – REACT

- Environment Setup
- Components
- JSX
- Props
- State
- Event Handling
- Conditional Rendering
- React Routing
- API Consumption

WEEK 10 – SPRING FRAMEWORK

- IoC and Dependency Injection
- Spring Container
- Maven Project Setup
- DAO Pattern
- Spring JDBC CRUD

WEEK 11 – SPRING BOOT

- Spring Boot Architecture
- Spring Data JPA
- RESTful CRUD APIs
- Exception Handling
- Spring Security Basics
- JWT Authentication

WEEK 12 – GIT AND GITHUB

- Git Basics
- Repository Creation
- Branching and Merging
- Conflict Resolution
- Reverting Commits
- GitHub Workflow